

Soc internship day 26

1. What is Threat Intelligence?

Threat Intelligence (TI) is the collection, analysis, and sharing of information about cyber threats to help organizations detect, prevent, and respond to attacks.

It answers questions such as:

- Who is attacking?
- What techniques are being used?
- Which systems are being targeted?
- How can we defend against the threat?

Types of Threat Intelligence

1. **Strategic Intelligence** – High-level information for management and decision-makers.
 2. **Tactical Intelligence** – Information about attacker tactics, techniques, and procedures (TTPs).
 3. **Operational Intelligence** – Details about ongoing attacks and threat actors.
 4. **Technical Intelligence** – Indicators of Compromise (IoCs) such as IP addresses, domains, URLs, and file hashes.
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2. Threat Intelligence Lifecycle Stages

1. Planning & Direction

- Define intelligence requirements.
- Identify assets and threats to monitor.

2. Collection

- Gather data from:
 - Security logs
 - Threat feeds
 - OSINT (Open Source Intelligence)
 - Dark web monitoring
 - Malware analysis

3. Processing

- Clean, organize, and normalize collected data.
- Remove duplicates and irrelevant information.

4. Analysis

- Convert raw data into actionable intelligence.
- Identify patterns, risks, and attacker behavior.

5. Dissemination

- Share intelligence with SOC analysts, incident responders, and management.

6. Feedback

- Evaluate effectiveness and refine future intelligence requirements.

Lifecycle Flow:

Planning → Collection → Processing → Analysis → Dissemination → Feedback

3. Malicious IP Scenario Analysis

Scenario

A company's firewall logs show repeated connections from IP address **203.0.113.10**.

Investigation Steps

Step 1: Identify the IP

- Check threat intelligence feeds.
- Search whether the IP is associated with:
 - Botnets
 - Malware
 - Command-and-Control (C2) servers

Step 2: Analyze Logs

Look for:

- Number of connection attempts
- Targeted systems
- Time pattern
- Ports used

Example:

Source IP: 203.0.113.10

Destination: Web Server

Port: 22 (SSH)

Attempts: 500 in 10 minutes

Step 3: Assess Risk

Indicators:

- Multiple failed login attempts
- Unusual geographic location
- Known malicious reputation

This suggests a **brute-force attack** against SSH services.

Step 4: Respond

- Block the IP on the firewall.
- Enable account lockout policies.
- Review authentication logs.
- Check for successful logins.
- Monitor for additional malicious activity.

Step 5: Update Intelligence

- Add the IP to internal blocklists.
- Share the IoC with the security team.
- Update detection rules in SIEM/EDR.

Outcome

The malicious IP is identified as part of a botnet performing SSH brute-force attacks. The organization blocks the IP, prevents unauthorized access, and improves monitoring.

Key IoCs in this Scenario

- Malicious IP address
- Repeated failed login attempts
- Unusual traffic patterns
- Suspicious ports and protocols

Threat intelligence helps organizations move from reactive defense to proactive defense by identifying threats before they cause damage.